Fiscal Year:	FY 2020	**Task Last Updated:**	FY 04/23/2021
PI Name:	Blanco, Gustavo Ph.D.		
Project Title:	Space Flight-Altered Motility Activation and Fertility-Dependent Responses in Sperm from Sea Urchin and Rodents		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	02/09/2018	**End Date:**	03/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 3/31/2021 per F. Hernandez/ARC (Ed., 4/7/2020) NOTE: Extended to 3/31/2020 per F. Hernandez/ARC and NSSC information (Ed. 5/15/19) NOTE: Extended to 8/31/2019 per F. Hernandez/ARC (Ed. 10/29/18)		
Key Personnel Changes/Previous PI:	January 2018: Former Principal Investigator was Joseph Tash, Ph.D., who retired January 2018. Dr. Tash remains as consultant on the project.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K0522		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (June 2018): Continuation of "Space Flight-Altered Motility Activation and Fertility-Dependent Responses in Sperm from Sea Urchin and Rodents" (grant number NNX13AM62G) due to retirement of former Principal Investigator Joseph Tash, Ph.D. Originally the grant had continued with same grant number as noted in January 2018 below note; in February 2018 new grant number 80NSSC18K0522 with new period of performance was awarded. NOTE (January 2018): Continuation of "Space Flight-Altered Motility Activation and Fertility-Dependent Responses in Sperm from Sea Urchin and Rodents" with the same grant number NNX13AM62G due to retirement of former Principal Investigator (PI) Joseph Tash, Ph.D. NOTE also that in 2016 with predecessor PI Tash, work on the grant was amended to include cryopreserved bovine and human semen, while eliminating the use of sea urchin and rat species. Sperm motility and acquisition of the capacity to fertilize are the most fundamental components of male reproduction necessary for species survival on Earth or in space. Previous space shuttle flight experiments that we performed demonstrated that the signal transduction (i.e., m protein phosphorylation) required for activation of sea urchin sperm occur more rapidly in microgravity than at 1G. In contrast, we found that signaling in sperm triggered by egg chemotactic factors is delayed in microgravity compared to 1G. No flight-capable microscope was available during those experiments to determine whether the signaling changes were in fact coupled to differences in motility under the microgravity versus 1G conditions. Our hypothesis is 1) that sperm motility (sea urchin and mammalian) is activated more rapidly in microgravity than on Earth, and 2) that the response of sea urchin and mammalian sperm (motility hyperactivation, and morphological changes) to fertilization activating conditions, as well as the underlying metabolic and regulatory mechanisms, are delayed or blocked in microgravity. In this regard, if the corresponding activation and changes in sperm motility do not occur, then fertilization cannot occur and reproduction will fail. Thus, it is essential to assess whether the critical component of sperm function, motility, and the underlying signalling and metabolic support are altered in space flight conditions. We propose to use existing flight hardware to activate sea urchin and rodent sperm in microgravity, and quantify multiple parameters of motility and morphology, combined with detailed metabolomic pathway analysis. Previously utilized methods for preparing inactive sea urchin sperm for subsequent activation in space flight will be adapted for mouse and rat sperm. We will also test whether activating peptides from sea urchin eggs, or mammalian capacitating conditions trigger temporal or magnitude changes in sperm motility, morphology, as well as essential regulatory glycolytic and energy metabolic pathway markers (using novel metabolomic analysis) in microgravity versus 1G. Most importantly, this comprehensive project will lead to the first direct in vitro assessment of canonical fertility-critical functional processes of mammalian sperm in microgravity that are identical to many human in vitro clinical and research laboratory assessments. These experiments are 1) part of the Rapid Turn-around Space Flight Experiments Research Emphasis, and 2) address the National Research Council (NRC) 2011 Decadal Survey to discover and characterize underlying mechanisms of adaptation to the space flight environment, and to determine cellular and organismal mechanisms that regulate and sustain reproduction and development in space. In this regard, the results of these experiments will provide important fundamental reproductive biological data to support NASA's goal to conduct multigenerational studies of mammals in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Previous studies indicate that reproductive functions in sperm that occur on the ground also occur in space flight. However, the timing of signal transduction changes upon initiation of sperm motility is more rapid, yet, the fertility-related responses of sperm to the presence of egg chemotactic factors are delayed. This suggests that sperm fertility may be affected resulting in subfertility or infertility in males during and post flight. There is insufficient data available from humans to assess this possibility due to low n's and HIPAA concerns.
Task Progress:	Reporting as of November 2020 Multi-generational survival of humans (and animals) require healthy functional sperm and eggs on Earth, in space flight, on potential habitats on the Moon, Mars, or beyond, and after return to Earth. On Earth, sperm function is routinely determined to assess male fertility and prior to applying any assisted reproduction technology (ART), including in vivo insemination (IUI), in vitro fertilization (IVF), or intracytoplasmic sperm injection (ICSI). Sperm capacity to fertilize eggs requires that the cells undergo critical changes (capacitation), which involve the acquisition of a specific pattern of sperm motility, and an exocytotic process in the sperm head called the acrosome reaction (AR). Quantitative assessments of motility and AR to live sperm on Micro-11 (SPX-14) using clinical procedures during space flight provided foundational data on prospects for long-term multigenerational survival in space. Triplicate cryopreserved human and bovine sperm samples (pooled from proven fertile 12 human and 6 fertile bulls, respectively) were thawed after 4 mo in orbit and activated for 60 min in IVF culture medium under capacitating and non-capacitating conditions. We reported previously that key motility biomarkers were significantly reduced/delayed in space flight sperm compared to asynchronous ground controls (GC) comprising triplicates of the same pools of human and bovine sperm. Here, we report that the rate of AR under IVF conditions was blunted in space flight compared to GC ($P<0.01$ by 3-way ANOVA) in human and bovine sperm. GC showed the expected increases in AR during the 60 min incubation under capacitating conditions, which were reduced under non-capacitating control conditions. There were no significant differences in the proportion of dead sperm in the space flight vs ground controls during the experiment. Our findings show that motility and acrosome reaction, both critical components of sperm required for successful fertilization, are significantly impaired under space flight condition. DNA integrity in sperm is also key for successful fertilization and to ensure that the proper genetic information is accurately passed on from one generation to the next. Any damage to the DNA may cause genetic instability, prevent successful fertilization, could be deleterious for the offspring, and risk multigenerational survival. Here, we assessed sperm DNA damage on our Micro-11 (SPX-14) cryopreserved proven fertile sperm pooled in triplicate from 12 proven fertile human donors. Samples were stowed frozen on board of the International Space Station (ISS) for 4 mo prior to thawing and activated in flight (FL) for 0-60 min under in vitro fertilization conditions as performed clinically on Earth. A replicate set of triplicate samples from the same aliquots of sperm served as asynchronous ground controls (GC). Sperm were fixed with PFA at 0 & 60 min activation, then returned to Earth and stained with Acridine Orange, a fluorescent biomarker for DNA damage. Quantitative fluorescent microscopy was used to determine the DNA Fragmentation Index (DFI %) in =500 cells per

	triplicate sample. Statistically significant increases (P==0.01) in DFI between human space flight and GC samples were observed both at 0 min (FL=16.0±1.4; GC DFI=8.0±1.4) and after 60 min activation (FL=32.7±1.7; GC=19.3±2.6). DFIs > 30 are high risk, 15-30 are moderate fertility risk. Thus, DNA damage increased during 4 mo stowage frozen, and during the 60 min incubation period compared to GC. These results suggest that human sperm DNA integrity, which is critical to fertility, is negatively impacted by space flight. While the origin of this DNA damage is unclear, two likely candidate factors for this event are the accumulation of space radiation and the effects of microgravity. Currently, we are performing similar determinations for bovine sperm to assess whether this is a phenomenon that occurs in another species. The data for humans raises questions as to the sperm fertilizing capacity for men in space.
Bibliography Type:	Description: (Last Updated: 04/23/2021)
Abstracts for Journals and Proceedings	Mitra A, Chancellor J, Zimmerman C, Feustel AJ, Aunon-Chancellor S, Yarns EM, Piper SS, Stodieck LS, Zea L, Doraisingam S, Karouia F, Blanco G, Tash JS. "Impact of Space Flight Radiation and Microgravity on DNA Integrity and Fertility Risk of Human & Bovine Sperm on the ISS." Presented at the 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. , Nov-2020
Abstracts for Journals and Proceedings	Tash JS, Mitra A, Feustel AJ, Aunon-Chancellor S, Yarns EM, Piper SS, Ngo L, Laundy E, Mittal V, Chanliogeo N, Julo M, Stodieck LS, Zea L, Doraisingam S, Karouia F, Blanco G. "Fertilization-Critical Motility & Acrosome Reaction of Human & Bovine Sperm are Negatively Impacted by Space Flight on the ISS." Presented at the 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. , Nov-2020
Abstracts for Journals and Proceedings	Mitra A, Chancellor J, Feustel AJ, Aunon-Chancellor S, Yarns EM, Piper SS, Stodieck LS, Zea L, Doraisingam S, Karouia F, Blanco G, Tash JS. "Impact of Space Radiation and Microgravity on Human Sperm DNA Integrity and Fertilizing Capacity." Presented at 2020 Greenwald Symposium, University of Kansas Medical Center, Kansas City, KS, October 8-9, 2020. 2020 Greenwald Symposium, University of Kansas Medical Center, Kansas City, KS, October 8-9, 2020. , Oct-2020
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